

Machine Learning — Algorithms, Models, and Best Practices

A detailed technical guide covering supervised learning, unsupervised learning, model evaluation, feature engineering, and modern ML practices.

1. Supervised Learning Algorithms

Supervised learning involves training a model on labeled input-output pairs. The model learns a mapping function from inputs (features) to outputs (labels/targets). This is the most widely applied form of ML in industry.

Linear Regression is used for predicting continuous numeric values. It fits a linear equation to minimize the sum of squared residuals. Variants include Ridge Regression (L2 regularization), Lasso (L1 regularization), and ElasticNet (combination of L1 and L2). Logistic Regression, despite its name, is a classification algorithm that models the probability of class membership using the logistic (sigmoid) function.

Decision Trees partition the feature space into rectangular regions using a series of binary splits on individual features. The tree is grown by choosing splits that maximize information gain (classification) or minimize variance (regression). Limitations include high variance and tendency to overfit. Ensemble methods address these limitations.

Support Vector Machines (SVMs) find the maximum-margin hyperplane that separates classes. The kernel trick allows SVMs to learn non-linear boundaries by implicitly mapping inputs to high-dimensional feature spaces. Common kernels include RBF (Radial Basis Function), polynomial, and linear. SVMs are effective in high-dimensional spaces and with limited training data.

2. Ensemble Methods

Random Forests are an ensemble of decision trees trained on bootstrapped samples of the data (bagging) with random feature selection at each split. Predictions are aggregated by majority vote (classification) or averaging (regression). Random Forests reduce variance compared to individual trees and provide feature importance estimates.

Gradient Boosting Machines (GBM) build an ensemble of weak learners (usually shallow trees) sequentially, where each tree corrects the errors of the previous ensemble. Popular

implementations include XGBoost, LightGBM, and CatBoost. XGBoost in particular dominated Kaggle competitions from 2014-2017 and remains a top performer on tabular data tasks.

Stacking (stacked generalization) combines predictions from multiple base models using a meta-learner. Blending is a simpler variant. Voting classifiers aggregate predictions from multiple models by majority vote (hard voting) or averaging probabilities (soft voting).

3. Unsupervised Learning

K-Means Clustering partitions data into K clusters by iteratively assigning points to the nearest centroid and recomputing centroids. The algorithm is sensitive to initialization — K-Means++ provides a smarter initialization strategy. Determining the optimal K can be done via the elbow method (plot inertia vs K) or silhouette analysis.

Hierarchical Clustering builds a tree of clusters (dendrogram) without requiring K to be specified in advance. Agglomerative (bottom-up) clustering starts with each point as its own cluster and merges clusters based on linkage criteria (single, complete, average, Ward's). DBSCAN clusters points by density and naturally identifies outliers as noise points.

Principal Component Analysis (PCA) is a dimensionality reduction technique that finds orthogonal directions of maximum variance. PCA is widely used for visualization (reducing to 2D/3D), noise reduction, and as preprocessing for downstream models. t-SNE and UMAP are non-linear dimensionality reduction methods better suited for visualization of high-dimensional data.

4. Neural Networks and Deep Learning

A neural network consists of layers of neurons (nodes) connected by weighted edges. The forward pass computes predictions; the backward pass (backpropagation) computes gradients of the loss with respect to all parameters. Stochastic Gradient Descent (SGD) and its variants (Adam, RMSProp, AdaGrad) update parameters in the direction that reduces loss.

Activation functions introduce non-linearity. ReLU (Rectified Linear Unit) is most commonly used in hidden layers due to its simplicity and effectiveness. Leaky ReLU and ELU address the 'dying ReLU' problem. Sigmoid and Tanh are used in output layers for binary classification and bounded regression respectively.

Batch Normalization normalizes layer activations during training, stabilizing learning and allowing higher learning rates. Dropout randomly zeroes neurons during training, acting as a regularizer. Residual connections (skip connections) in ResNet architectures allow gradients to flow through very deep networks.

5. Model Evaluation and Validation

For classification tasks, key metrics include Accuracy (fraction of correct predictions), Precision (fraction of positive predictions that are correct), Recall/Sensitivity (fraction of actual positives correctly identified), F1 Score (harmonic mean of precision and recall), and AUC-ROC (area under the Receiver Operating Characteristic curve).

For regression tasks, common metrics are Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (coefficient of determination). The choice of metric should align with the business objective — e.g., RMSE penalizes large errors more than MAE.

K-Fold Cross Validation partitions data into K folds, trains K models each on K-1 folds, and evaluates on the held-out fold. Results are averaged for a robust performance estimate. Stratified K-Fold maintains class proportions in each fold. Leave-One-Out CV is the extreme case where K equals the number of samples.

6. Feature Engineering

Feature engineering is the process of using domain knowledge to create, transform, or select features that improve model performance. For numeric features: scaling (StandardScaler, MinMaxScaler), log transformation for skewed distributions, polynomial features for capturing non-linear relationships. For categorical features: one-hot encoding, ordinal encoding, target encoding.

Missing value imputation strategies: mean/median imputation (simple but ignores correlations), KNN imputation (uses nearest neighbors), iterative imputation (models each feature as a function of others). Outlier handling: detection via IQR or Z-score, then removal or capping. Feature selection methods: filter methods (correlation, chi-square), wrapper methods (recursive feature elimination), embedded methods (Lasso, tree-based importance).

7. Hyperparameter Tuning

Hyperparameters are model settings not learned from data (e.g., learning rate, number of layers, regularization strength). Grid Search exhaustively searches a specified parameter grid. Random Search samples random combinations and is often more efficient. Bayesian Optimization builds a probabilistic model of the objective function and selects the next hyperparameters to evaluate intelligently.

Modern AutoML systems (Auto-sklearn, H2O AutoML, Google AutoML) automate the entire pipeline — preprocessing, model selection, and hyperparameter tuning. Neural Architecture Search (NAS) automates the design of neural network architectures, discovering architectures that outperform human-designed ones.

8. ML in Production (MLOps)

MLOps is the practice of deploying and maintaining ML models in production reliably and efficiently. Key concerns include model versioning (MLflow, DVC), feature stores (Feast, Tecton), model serving (FastAPI, TorchServe, Triton), monitoring for data drift and model degradation, and automated retraining pipelines.

CI/CD for ML extends traditional software CI/CD to include data validation, model training, evaluation, and deployment steps. Containerization (Docker) and orchestration (Kubernetes) enable scalable, reproducible ML deployments. A/B testing and shadow mode deployment allow safe rollout of new model versions.